INCH-POUND

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PERFORMANCE SPECIFICATION

FILTER/MONITOR, CONTAMINATION, HELICOPTER IN-FLIGHT REFUELING SYSTEM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers aviation fuel contamination filter/monitor elements only for existing Helicopter In-Flight Refueling (HIFR) housings and should not be used for new construction or existing ground handling systems.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center, Aircraft Division Lakehurst, Code 4.1.2, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-PRF-25017	-	Inhibitor, Corrosion/Lubricity Improver, Fuel Soluble, (NATO-S1747)
MIL-DTL-83133	-	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)
MIL-DTL-85470	-	Inhibitor, Icing, Fuel System, High Flash NATO Code Number S-1745

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests

NORTH ATLANTIC TREATY ORGANIZATION (NATO) STANDARD

AFLP-3747	-	Guide Specifications (Minimum Quality Standards) for Aviation Turbine Fuels (F-24, F-27, F-34, F-35, F-37, F-40 and F-44)

(Copies of these documents are available online at http://quicksearch.dla.mil/ or https://assist.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ-Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted)

(Copies of this document are available from www.asq.org.)

ASTM INTERNATIONAL

ASTM D910	-	Standard Specification for Leaded Aviation Gasolines
ASTM D1331	-	Standard Test Methods for Surface and Interfacial Tension of Solutions of Surface and Interfacial Tension of Solutions of Paints, Solvents, Solutions of Surface-Active Agents, and Related Materials
ASTM D1655	-	Standard Specification for Aviation Turbine Fuels
ASTM D2276	-	Standard Test Method for Particulate Contaminant in Aviation Fuel by Line Sampling

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ASTM D3240	-	Standard Test Method for Undissolved Water in Aviation Turbine Fuels
ASTM D5006	-	Standard Test Method for Measurement of Fuel System Icing Inhibitors (Ether Type) in Aviation Fuels
ASTM D7224	-	Standard Test Method for Determining Water Separation Characteristics of Kerosine-Type Aviation Turbine Fuels Containing Additives by Portable Separometer

(Copies of these documents are available from www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

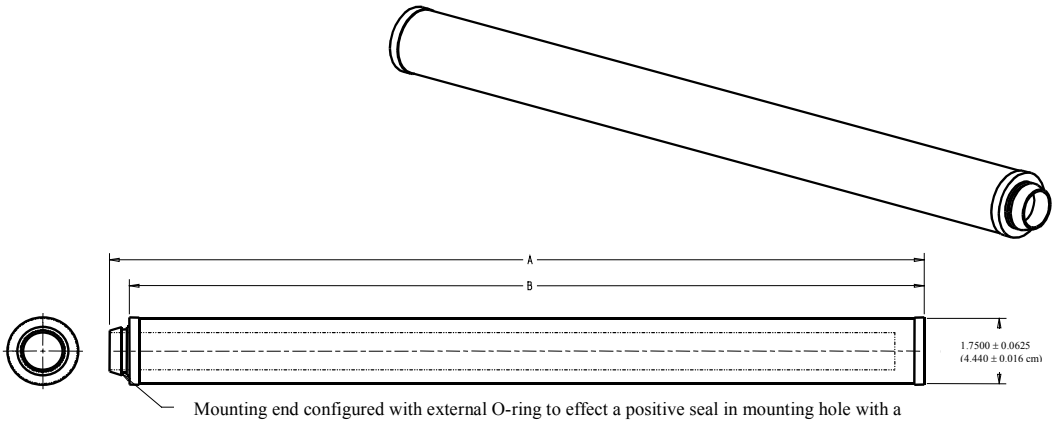
3. REQUIREMENTS

3.1 Qualification. The fuel monitor elements furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3). Qualification testing may be performed with written authorization from the qualifying activity.

3.2 Material compatibility. Materials used in aviation fuel monitors shall not be adversely affected by and shall have no effect on fuels and inhibitors conforming to MIL-DTL-5624, MIL-PRF-25017, MIL-DTL-83133, MIL-DTL-85470, ASTM D910, ASTM D1655, and F-24 as specified in AFLP-3747. Compatibility with additional fuels and inhibitors shall be specified in the contract if necessary (see 6.2).

3.3 Element configuration and flow rate. The filter/monitor elements shall be as shown on Figure 1. Thirty-inch nominal length elements shall be used for qualification testing. Ten-inch and 20-inch nominal length elements shall be procured by the contracting activity for HIFR (see 6.2). Direction of flow shall be from outside to inside of the element. The open end shall be configured so as to effect a positive, non-leaking seal when seated properly in the filter/monitor housing. Each element shall have a minimum flow rate of one gallon per minute (gpm) per inch of nominal length (0.063 liter per second per 2.54 centimeters). The filter/monitor element shall not include any device that will cause an immediate or sudden stoppage of fuel flow.

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NOMINAL LENGTH	FLOW RATING	A	B
30 (76.2 cm)	30 GPM (114 LPM)	30.8125 ± 0.0625 (78.26 ± 0.16 cm)	30.250 ± 0.0625 (76.64 ± 0.16 cm)
20 (50.8 cm)	20 GPM (76 LPM)	20.8125 ± 0.0625 (52.86 ± 0.16 cm)	20.250 ± 0.0625 (51.44 ± 0.16 cm)
10 (25.4 cm)	10 GPM (38 LPM)	10.8125 ± 0.0625 (27.48 ± 0.16 cm)	10.250 ± 0.0625 (26.04 ± 0.16 cm)

Notes:
1. Unless otherwise specified, dimensions are in inches.
2. All materials shall be compatible with aviation fuels (see 3.2 and 6.2)
3. Direction of flow is outside to inside.

FIGURE 1. Monitor element.

3.4 Performance characteristics.

3.4.1 Structural integrity. The filter/monitor element shall withstand a three-foot (91.4-cm) drop without structural failure, when tested in accordance with 4.5.2.1.

3.4.2 Media migration. The effluent shall contain an average of not more than 10 fibers per liter (see 6.6.3) and the number of fibers in any single sample shall not be greater than 15 fibers per liter, when tested in accordance with 4.5.2.1.

3.4.3 Differential pressure. With a new element installed in a filter/monitor housing (single element test), the differential pressure across the element shall not be greater than 8 psi (55.15 kPa) at rated flow (see 3.3), when tested in accordance with 4.5.2.1.

3.4.4 Reaction to solids. The average of total solids in the effluent fuel allowed by the filter/monitor shall not be greater than 0.26 mg/liter. The average total solids shall consist of the combined tare weights of the individual sample filters [membrane filters used to measure the solids in the effluent stream (see 4.4.1.4)] divided by the total volume of fuel passed through the samples. No individual sample shall exceed 0.5 mg/liter. Solids content shall be tested in

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accordance with ASTM D2276. The time to reach a differential pressure of 75 psi (517.05 kPa) shall not be less than 10 minutes at 100 percent of rated flow, when tested in accordance with 4.5.2.2.

3.4.5 Mechanical integrity. The filter/monitor element shall withstand a pressure differential of 180 psi (1241.1 kPa) without deformation, when tested in accordance with 4.5.2.2.

3.4.6 Reaction to water. The filter/monitor element's reaction to water shall be as shown in Table I, when tested in accordance with 4.5.2.3.1. The minimum running time to 25 psi (172.4 kPa) differential pressure shall be as specified in Table I for test conditions 1 and 2. When analyzed in accordance with ASTM D3240, the effluent shall not contain more than 10 parts per million (ppm) by volume of free water.

TABLE I. Filter/monitor element's reaction to water.

Test Condition	Flow Rate (%)	Running Time (minutes)	Free Water Content (by volume)		Total Leakage (% of Flow Rate)	Sampling Intervals (minutes)
			Influent	Effluent		
1	100	15 (minimum)	50 ppm	10 ppm (max)	-	5
2	10	100 (minimum)	50 ppm	10 ppm (max)	-	60
3	100	-	1000 ppm Water in Fuel (Slug)	-	5.0%	1
4	100	-	50% Water 50% FSII (Slug)	-	5.0%	1
5	100	-	100% Water (Slug)	-	5.0%	1

3.4.7 Reaction to surge. The filter/monitor element shall not pass water in excess of 10 ppm or be otherwise adversely affected by a surge of fuel, when tested in accordance with 4.5.2.3.1.

3.4.8 Reaction to slugs. When triggered by 1000 ppm water in fuel, a 50/50 water/FSII slug, or 100 percent water slug (conditions 3, 4, and 5 of Table I, respectively), the total fluid leakage through the device shall not exceed 5.0 percent of the rated flow to the device, when tested in accordance with 4.5.2.3.2.

3.4.9 Reaction to water immersion. The fuel filter/monitor elements shall show no evidence of excessive bulging, rupture or media extrusion, which could cause failure during operation after being immersed in water conforming to 4.4.4.1. The total fluid leakage through the device shall not exceed 5.0 percent of the rated flow to the device, when tested in accordance with 4.5.2.4.
NOTE: Excessive bulging is determined to be any increase of more than ¼ inch total diameter.

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3.4.10 Reaction to freezing/thawing. After the element has been frozen and then thawed in accordance with 4.5.2.5, the filter/monitor shall not pass water in excess of 10 ppm.

3.4.11 Performance after extended flow test. Using fuel conforming to 4.4.3, the filter/monitor element shall not pass water in excess of 10 ppm, when tested in accordance with 4.5.2.6.

3.4.12 Environmental requirements.

3.4.12.1 High temperature. The fuel filter/monitor elements shall withstand storage without deterioration in temperatures as high as 160 °F (71 °C), when subjected to the test specified in 4.5.3.1. They shall also permit rated flow in ambient temperatures as high as 125 °F (52 °C) (see 6.2 and 6.5).

3.4.12.2 Low temperature. The fuel filter/monitor elements shall withstand storage without deterioration in temperatures as low as -50 °F (-46 °C) when subjected to the test specified in 4.5.3.2. They shall also permit rated flow in ambient temperatures as low as -25 °F (-32°C) (see 6.2 and 6.5).

3.4.12.3 Fuel immersion. The filter/monitor element shall show no evidence of swelling, corrosion, separation of components, dissolving of adhesives, or deformation, which could cause failure during operation after being immersed in additized fuel conforming to section 4.4.3 and additized in accordance with 4.4.3.3, and tested in accordance with 4.5.3.3.

3.4.13 Post-environmental. The filter/monitor elements subjected to the conditions of 3.4.12.1, 3.4.12.2, and 3.4.12.3 shall operate without failure, when tested in accordance with 4.5.4. The effluent shall contain no more than 10 ppm by volume free water as determined in accordance with ASTM D3240.

3.5 Workmanship. Elements shall be free of burrs and sharp edges. They shall be uniform in quality and shall be free from irregularities or defects that could adversely affect performance or durability.

3.5.1 Marking. Filter/monitors qualified to this specification shall be permanently marked with model/inventory number 81380F-XX where X is the nominal length of the element in inches. The model/inventory number shall be used as the sole identification to identify filter monitors qualified to this specification. Additionally, each element shall be marked with the following information:

a. Manufacturer name
b. Contract or order number
d. Lot number
e. National Stock Number (NSN)
f. Date of manufacture

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 Qualification inspection. The primary qualification inspection shall be performed using elements with a nominal length of 30 inches (76.2 cm) and shall consist of the examinations and performance, environmental, and post-environmental tests listed in Table II. Ten-inch and 20-inch nominal length elements shall be qualified by similarity using the examination criteria listed in Table III. Five 10-inch elements and five 20-inch elements shall be examined. Only elements that have passed qualification inspection shall be listed on the QPL.

TABLE II. Filter/monitor element qualification testing.

Inspection	Requirement Paragraph	Method Paragraph
Examination	-	-
Material Compatibility	3.2	4.5.1
Element Configuration and Flow Rate	3.3	4.5.1
Workmanship	3.5	4.5.1
Performance, environmental, and post-environmental tests	-	-
Structural integrity, media migration, and differential pressure test	3.4.1, 3.4.2, 3.4.3	4.5.2.1
Solids reaction and mechanical integrity test	3.4.4, 3.4.5	4.5.2.2
Water and surge reaction test	3.4.6, 3.4.7	4.5.2.3.1
Reaction to slugs test	3.4.8	4.5.2.3.2
Water immersion test	3.4.9	4.5.2.4
Freezing/thawing test	3.4.10	4.5.2.5
Performance after extended flow test	3.4.11	4.5.2.6
High temperature test	3.4.12.1	4.5.3.1
Low temperature test	3.4.12.2	4.5.3.2
Fuel immersion test	3.4.12.3	4.5.3.3
Post-environmental water reaction test	3.4.13	4.5.4

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TABLE III. Supplemental filter/monitor element qualification testing.

Inspection	Requirement Paragraph	Method Paragraph
Examination	-	-
Material Compatibility	3.2	4.5.1
Element configuration and flow rate	3.3	4.5.1
Workmanship	3.5	4.5.1

4.2.1 Qualification inspection filter/monitor element samples. Primary qualification samples shall consist of 40 elements with a nominal length of 30 inches (76.2 cm). Five 10-inch (25.4-cm) elements and five 20-inch (50.8-cm) elements shall be submitted for qualification inspection.

4.2.2 Retention of qualification. Retention of qualification for filter/monitor elements on the QPL shall be dependent on periodic verification, by means of test data, of continued compliance with the requirements of this specification. This periodic verification shall be conducted every five years from the date of inclusion on the QPL and will consist of the testing of forty 30 inch elements in the same manner as an initial qualification. Continued qualification of 10-inch and 20-inch elements shall be determined through examination testing of five of each length element (see Table III) and verification testing of 30 inch elements. These qualification retention tests will be performed by the qualifying activity (see 6.3). Complete re-qualification may be required if any change or substitution is made in the materials or manufacturing process of the elements. The qualifying activity may grant written waivers for minor changes in materials or processing provided a risk assessment determines the changes will have negligible impact on element or system performance.

4.3 Conformance inspection.

4.3.1 Conformance inspection for elements. Conformance inspection for elements shall consist of the tests shown in Table IV. These inspections shall be performed on those samples selected from an inspection lot in accordance with ASQ-Z1.4, general inspection level II. An inspection lot shall consist of all elements of the same size produced under the same conditions and offered for inspection at one time.

TABLE IV. Group A conformance inspection for elements.

Inspection	Requirement Paragraph	Method Paragraph
Examination		-
Material Compatibility	3.2	visual inspection
Workmanship	3.5	visual inspection

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4.4 Inspection conditions.

4.4.1 Equipment. Equipment used for testing shall be of laboratory precision type or equivalent as approved by the qualifying activity insofar as practicable, and shall be calibrated at intervals properly spaced to ensure laboratory accuracy. Figure 2 is a schematic of a filter/monitor qualification test system. At a minimum, all equipment shown on Figure 2 shall be present in a filter/monitor qualification test system.

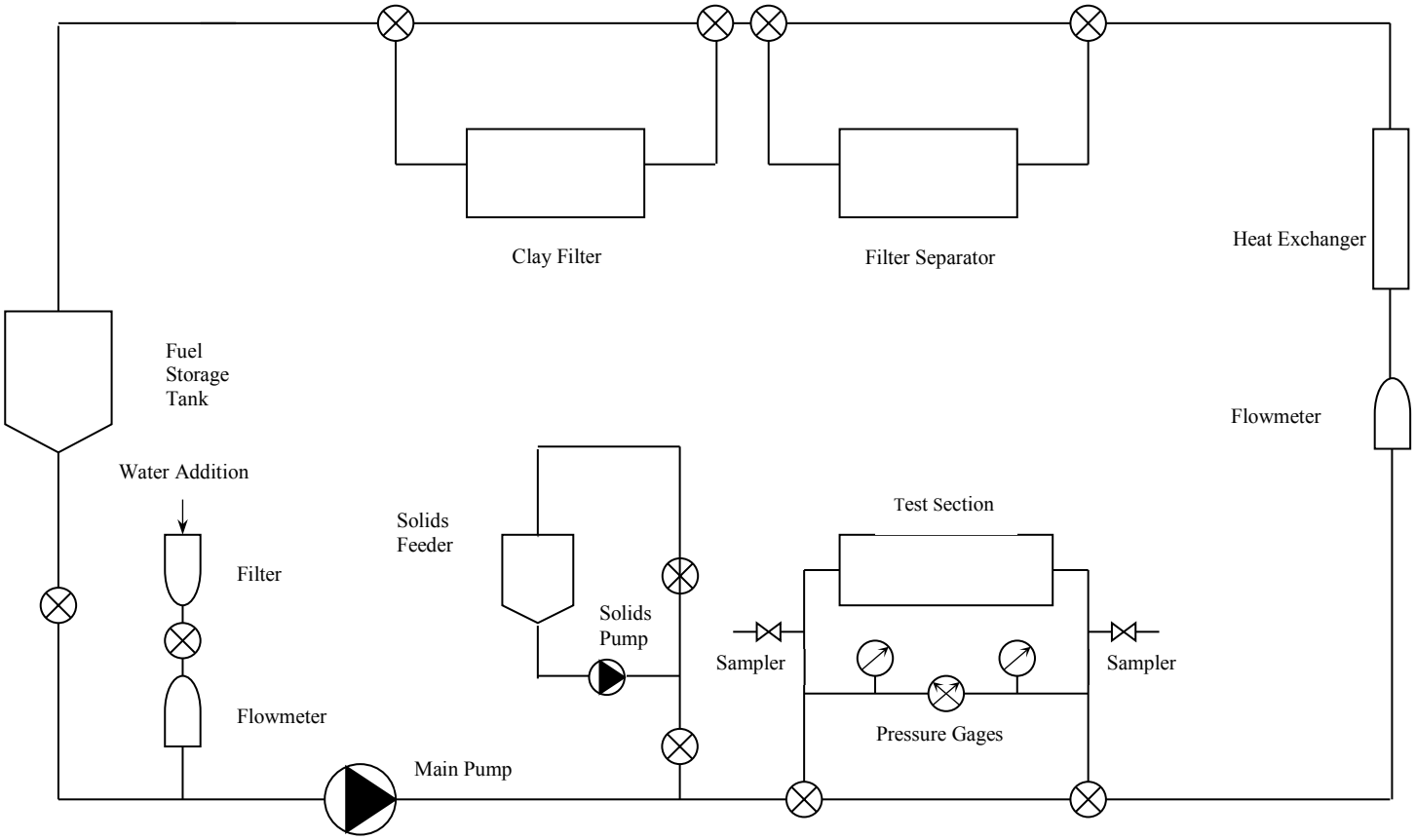


FIGURE 2. Test system schematic.

4.4.1.1 Water injection system. A system shall be provided to inject water into the fuel prior to the main pump. The accuracy of the water injection system shall be within five percent of the specified water addition rate.

4.4.1.2 Main circulation pump. The main circulating pump(s) shall be a centrifugal or vane pump capable of emulsifying the fuel and water mixture. A minimum pump speed of 2500 rpm shall be maintained to ensure consistent emulsification.

4.4.1.3 Solids addition system. A continuous solids addition system shall be provided to uniformly feed test dust at a rate of 10 mg/1iter into the fuel upstream of the test unit.

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4.4.1.4 Sampling devices. Upstream-facing, probe-type sampling devices shall be provided, one immediately before and one immediately after the test unit, for extracting in-line samples and passing them through membrane or uranine dye-treated filters.

4.4.1.5 Differential pressure reading device. Differential pressure across the test unit shall be measured with an accuracy of two percent of the scale reading of the calibrated (see 6.7) device used, such as a differential pressure gage, selector valve and gage, etc.

4.4.1.6 Pressure relief valve. An adjustable pressure relief valve shall be installed before the test unit for system protection.

4.4.1.7 Fuel flow meter. A calibrated (see 6.7) flow meter shall be used to measure the fuel flow with an accuracy of one percent at the specified fuel flow rate.

4.4.2 Test system pressure. The pressure in the test system shall be not less than 20 psi (137.8 kPa) at the start of all tests when measured at the sampler immediately downstream of the test unit.

4.4.3 Test fluid. Unless otherwise specified, the test fluid shall be aviation fuel that conforms to MIL-DTL-5624, MIL-DTL-83133, ASTM D1655, or F-24 as specified in AFLP-3747 and contains all of the additives listed in 4.4.3.3. The fluid shall be clean fuel (see 6.6.1) and shall be tested for specification conformance

4.4.3.1 Preliminary testing. Prior to the start of testing, the fuel shall be evaluated for conformance to the applicable specification.

4.4.3.2 Fuel pre-treatment. Before initiating any filter/monitor performance testing, the fuel shall be subjected to clay treatment until a Micro-Separometer (MSEP) Surfactants value of 95 or greater is obtained, when tested in accordance with ASTM D7224 and have a Fuel System Icing Inhibitor concentration less than 0.01 percent, when tested in accordance with ASTM D5006.

4.4.3.3 Required additives and their injection. The procedure outlined in Appendix A shall be used to inject the test fuel with the following additives to the specified concentrations:

a. Static Dissipater Additive, Stadis 450® (Octel America, Newark DE) at a concentration of 2.0 mg/liter, ±0.2 mg/liter.

b. Fuel System Icing Inhibitor in accordance with MIL-DTL-85470 at a concentration of 0.11 percent (by volume), ±0.01 percent.

c. Corrosion Inhibitor DCI-4A in accordance with MIL-PRF-25017 at a concentration of 15 mg/liter, ±1 mg/liter.

4.4.3.4 Test fuel volume. Test fuel flow shall be single pass (i.e., non-recirculating) for single element tests. The qualifying activity may grant written waivers to perform any portion of this qualification with recirculating fuel flow.

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4.4.3.5 Fuel temperature. The test fluid temperatures shall be maintained between 70 °F and 90 °F (21 °C and 32 °C) for all tests specified herein, and shall be kept within ±5 °F (±3 °C) for the duration of the test. .

4.4.4 Test contaminants. The following fresh water and solids shall be used.

4.4.4.1 Fresh water. Fresh water used for testing purposes shall be filtered and contain less than 1.0 mg/liter of solids and shall have a surface tension of not less than 65 dynes per centimeter at 75 °F (24 °C) when tested in accordance with ASTM D1331, Method A. The pH value shall be between 6.0 and 8.0.

4.4.4.2 Solids. The solids test mixture shall consist of 10 percent by weight of Copperas Red Iron Oxide, R9998 and 90 percent A1 Ultrafine ISO 1201-1 test dust. Alternate test dusts may be used with prior written authorization from qualifying activity.

4.5 Inspection methods.

4.5.1 Examination. Each element selected (see 4.3.1) shall be examined for conformance to the requirements listed in Tables II, III, and IV as applicable. Visual inspections shall be performed before and after all tests to determine conformance with the requirements of 3.2, 3.3, and 3.5.

4.5.2 Performance tests.

4.5.2.1 Structural integrity, media migration, and differential pressure test. Three elements shall be dropped from a distance of three feet above a concrete surface with each element in a vertical position, open end down. Three feet shall be measured from the concrete surface to the bottom open end of the element while in a vertical position. Visual inspection shall be performed on the elements to identify cracking of the material or other structural defects (see 3.4.1). The same elements shall then be tested in a single element test facility as shown on Figure 2. A test fluid conforming to 4.4.3 and additized in accordance with 4.4.3.3 shall be pumped through the test unit for a period of 15 minutes using the flow schedule shown in Table V. One 1000 mL sample shall be taken from the inlet sampler and one 1000 mL sample shall be taken from the outlet sampler immediately after starting at rated flow. This sampling procedure shall be repeated during the transition from rated flow to 10 percent of the rated flow and during the transition period back to 100 percent rated flow. A fiber count of the samples for media migration shall be conducted in accordance with Appendix B. The media migration result shall meet the requirements of 3.4.2. In addition, the differential pressure across the element shall be recorded at the start and finish of each phase of Table V. The differential pressure shall meet the requirements of 3.4.3.

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TABLE V. Flow schedule.

Phase	Elapsed Time (minutes)	Flow Rate (% rated flow)
1	0 - 5	100
2	5 - 10	10
3	10-15	100

4.5.2.2 Solids reaction and mechanical integrity test. Using the three elements that were tested in 4.5.2.1, each element one by one shall be tested in the single element test vessel at 100 percent of rated fuel flow. Test fuel conforming to 4.4.3 shall be used. Contaminant conforming to 4.4.4.2 shall be injected at a rate of 10 mg/liter into the fuel stream. The test dust shall be continuously and uniformly added to the fuel influent stream as close as practicable to the test housing. The test shall be continued until the differential pressure across the element reaches 180 psi (1241.1 kPa) and then held at 180 psi for five minutes. Continuous samples up to a pressure differential of 25 psi shall be taken at the outlet samplers and analyzed for solids content in accordance with ASTM D2276. These sampler membrane filters may be replaced as they become loaded with solid contaminants. The results of solids testing shall meet the requirements of 3.4.4. The differential pressure shall be recorded every five minutes and the results shall meet the requirements of 3.4.5. An alternate method of completing this test is to remove the element after 75 psi is achieved and placing it in a second system capable of higher pressures.

4.5.2.3 Water, surge and slug reaction tests.

4.5.2.3.1 Water and surge reaction test. Using a new element for each test, six separate single element tests shall be conducted in accordance with conditions 1 and 2 of Table I (three tests for each condition). Test fuel conforming to 4.4.3 shall be used. Water conforming to 4.4.4.1 shall be injected into the fuel on the inlet side of the main pump at the rates shown for conditions 1 and 2 of Table I. The differential pressure shall be recorded every five minutes. Sampling shall be done at the intervals specified in table I and analyzed in accordance with ASTM D3240. When the differential pressure reaches 15 psi (103.35 kPa), the fuel flow shall be momentarily stopped and restarted by closing and, within four seconds, opening the downstream valve. The main pump shall be kept running during the cycling of the valves to provide a surge of fuel. The surge of fuel shall be repeated when the differential pressure reaches 20 psi (137.8 kPa). Immediately after initiating a surge (closing and reopening the downstream valve), an effluent free water measurement shall be conducted in accordance with ASTM D3240. Effluent samples collected after fuel surges shall meet the requirements specified in 3.4.7. The test shall be continued until the pressure differential reaches 25 psi (172.25 kPa). Effluent free water concentrations shall meet the requirements of 3.4.6 and 3.4.7.

4.5.2.3.2 Reaction to slugs. Using a new element for each test, nine separate single element tests shall be conducted in accordance with conditions 3, 4, and 5 of Table I (three tests for each condition). Test fuel conforming to 4.4.3 shall be used. Water shall meet the requirements of 4.4.4.1. The minimum slug volume shall be 1 percent of the element's rated flow rate or equivalent to the volume of slug containing the water holding capacity of the element (varies based on test condition). The greater volume of the two scenarios shall be used. The maximum

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volume of water used shall be equivalent to the volume of pipe work between the water slug injection point and the test vessel plus the test vessel volume upstream of the elements. This is the volume of water to be used in the 50/50 slug mixture. To this volume, add the same volume of FSII. Stir the mixture well in a container. Use caution when mixing since the mixture will become warm. The same volume of water shall be used for the 100 percent water slug as the total volume of the 50/50 water/FSII slug. Following slug introduction, fuel shall back the slug.

Test conditions 3, 4, and 5 of Table I shall be completed as follows:

1. Circulate fuel conforming to 4.4.3 and additized in accordance with 4.4.3.3 through the test vessel and element at full rated flow.
2. Keeping the pump running, smoothly introduce the fuel/1000 ppm water slug (condition 3) upstream of the pump using the pump suction. The 50/50 water/FSII slug (condition 4) and 100 percent water slug (condition 5) may be injected upstream of the pump or downstream of the pump by diverting the fuel through a side stream containing the prerequisite volume of slug.
3. Flow shall be continued until the pump reaches stall pressure. The stall pressure shall be maintained for five minutes and the effluent from the test vessel shall be collected into separate graduated containers. One container for each minute shall be used. The leak rate of the effluent fuel shall meet the requirements of 3.4.8. To accomplish this test, a downstream valve shall be shut off to not allow flow back to the fuel tank. The leak rate sample shall be taken from a probe or sample point on the downstream side of the test housing. The first 5 seconds of leak by flow may be discarded to allow for the depressurizing of the system.

4.5.2.4 Water immersion test. A new element shall be fully immersed in water conforming to 4.4.4.1 for a minimum of 10 hours. The inlet shall be plugged to ensure that no water enters the downstream side of the element. It shall then be removed from the water and visually inspected. Any evidence of excessive bulging, rupture, or media extrusion shall be cause for failure. It shall then be installed in the test vessel and exposed to at least 75 psi (517.1 kPa) pressure by flowing fuel conforming to 4.4.3 through the vessel. The leak rate sample shall be determined by collecting samples from a probe or sample point downstream of the test housing each minute for five minutes. Each sample shall be collected into a separate graduated container. The effluent fuel shall meet the requirements of 3.4.9.
NOTE: Excessive bulging is determined to be any increase of more than ¼ inch total diameter.

4.5.2.5 Freezing/thawing test. Two separate single element tests shall be conducted using a new element and fuel conforming to 4.4.3 and additized in accordance with 4.4.3.3 for each test. An element shall be installed in the single element test vessel and test condition 1 of table I shall be run in accordance with 4.5.2.3.1 until a differential pressure of 15 psi (103.4 kPa) is reached (a surge at 15 psi is not required). The element shall be removed and placed in a bath of fuel conforming to 4.4.3 and additized in accordance with 4.4.3.3. The temperature of the bath shall be lowered and maintained at a maximum temperature of -25 °F (-32 °C) for 24 hours. The element shall then be returned to ambient condition, reinstalled in the test vessel and subjected to the remainder of test condition 1 of Table I in accordance with 4.5.2.3.1 (a surge at 15 psi and 20 psi is required). Effluent sample results shall meet the free water requirements of 3.4.10. Surge

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reaction shall meet the requirements of 3.4.7. This procedure shall be repeated using a second new filter/monitor element.

4.5.2.6 Performance after extended flow test. Three separate single element tests shall be conducted. A minimum of 5,000 gallons of fuel conforming to 4.4.3 and additized in accordance with 4.4.3.3 shall be circulated through the element at 100 percent rated flow for 24 hours, bypassing all clean-up units. The elements may be installed in a multi-element housing for the 24-hour conditioning period. After the 24-hour conditioning period, a water and surge reaction test using test condition 1 of Table I shall be conducted in accordance with 4.5.2.3.1. Effluent sample results shall meet the free water requirements of 3.4.11. Surge reaction shall meet the requirements of 3.4.7.

4.5.3 Environmental tests. Two new elements shall be subjected to the tests specified in 4.5.3.1, 4.5.3.2, and 4.5.3.3. The same elements shall be used for all three tests. Those elements shall then be tested in accordance with 4.5.4.

4.5.3.1 High temperature test. Two elements shall be tested in accordance with MIL-STD-810 Method 501.6 Procedure I steps 1, 2, and 3b, using a constant temperature of 160 °F (71 °C) for 24 hours. The test items shall then be examined for defects. (See 3.4.12.1.)

4.5.3.2 Low temperature test. The elements tested in 4.5.3.1 shall be tested in accordance with MIL-STD-810, Method 502.6 Procedure I steps 1, 2, 3, and 6, using a temperature of -50 °F for 24 hours. The test items shall then be examined for defects. (See 3.4.12.2.)

4.5.3.3 Fuel immersion test. The two elements tested in 4.5.3.1 and 4.5.3.2 shall be immersed in fuel conforming to 4.4.3 for 100 hours at a temperature of 77 ±5 °F (25 ±3 °C). The elements shall then be removed and examined to determine compliance with 3.4.12.3.

4.5.4 Post-environmental water reaction test. After successful completion of the tests specified in 4.5.3.1, 4.5.3.2, and 4.5.3.3, the same two elements shall be, one at a time, installed and tested in the test vessel. A water reaction test shall be conducted in accordance with 4.5.2.3.1 (surge reaction testing not required), using test condition 1 of Table I.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's System Commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

6.1 Intended use. This specification covers aviation fuel contamination filter/monitor elements for existing Helicopter In-Flight Refueling (HIFR) housings only and is not to be used for new construction or existing ground handling systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Whether materials are to be compatible with additional fuels and inhibitors (3.2).

c. Element length (3.3).

d. Mechanical integrity test results (see 3.4.5, 4.5.2.2, and 6.4).

e. Rated flow test results (see 3.4.12.1, 3.4.12.2 and 6.5).

f. Identification markings (see 3.5.1).

g. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-81380, whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Contractors may perform qualification testing with prior written authorization from the qualifying activity. Information pertaining to qualification of products may be obtained from Naval Air Systems Command, Fuels and Lubricants Division, Building 2360, 22229 Elmer Road, Patuxent River, MD 20670-1534. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Thirty-inch nominal length elements qualification. Primary qualification testing must be completed using 30-inch nominal length elements unless the NAVAIR qualifying activity specifies otherwise, in which case an alternate length monitor must be provided for qualification testing. Manufacturer provided mechanical integrity and rated flow test results may be from tests completed with 30-inch, 20-inch, or 10-inch length monitors (see 6.4 and 6.5).

6.3.2 Ten-inch and 20-inch nominal length element qualification. Ten-inch and 20-inch nominal length elements are qualified by similarity using the examination criteria listed in Table III.

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6.3.3 Test fuel specification conformance. Manufacturers requesting qualification approval must provide a data sheet proving the test fuel conforms to its applicable specification to the NAVAIR qualifying activity.

6.4 Mechanical integrity test results. Upon request, manufacturer must provide monitor mechanical integrity data confirming elements withstand a differential pressure of 180 psi for 5 minutes without deformation (see 3.4.5 and 4.5.2.2).

6.5 Rated flow test results.

6.5.1 Rated flow in high temperatures. Upon request, manufacturer must provide temperature flow data confirming elements permit rated flow in ambient temperatures as high as 125 °F.

6.5.2 Rated flow in high temperatures. Upon request, manufacturer must provide temperature flow data confirming elements permit rated flow in ambient temperatures as low as -25 °F.

6.6 Definitions.

6.6.1 Clean fuel. Clean fuel (test fluid) is defined as fuel containing less than 0.26 mg/liter of solids contamination and less than 5 parts per million (ppm) free water.

6.6.2 Differential pressure. The differential pressure across the aviation fuel contamination filter/monitor is defined as the pressure drop measured from the fuel inlet connection to the fuel outlet connection.

6.6.3 Fiber. A fiber is defined as any particle with a length-to-diameter ratio of 10 or greater and a minimum length of 100 microns.

6.7 Calibrated equipment. A calibration certificate must be made available upon request.

6.8 Subject term (key word) listing.

Aircraft
Gasoline
Impurity
Sensor

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

METHOD FOR INJECTING FUEL ADDITIVES TO THE TEST FLUID

A.1 SCOPE

A.1.1 Scope. This appendix details the method to be used for injecting the test fluid with the required additives. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 PROCEDURE

A.2.1 Test fuel cleanup. The test fluid shall be filtered until an AquaGlo reading of 5 ppm by volume or less is obtained when tested in accordance with ASTM D3240. The test fluid shall then be water washed and clay filtered until a Micro-Separometer (MSEP) Surfactants value of ≥95 and a FSII concentration of 0.01 percent is obtained when tested in accordance with ASTM D7224 and ASTM D5006, respectively. All filtration equipment shall be bypassed before adding the additives.

A.2.2 Inhibitor injection. The test fuel shall be inhibited according to the amounts specified in section 4.4.3.3. To determine the duration of recirculation needed to achieve a homogenous mixture of fuel and inhibitors, the following procedure shall be used. Inject additive A upstream of the main pump. The conductivity shall be measured at 5 minute intervals after the additive is introduced to the fuel. The elapsed time from the initial addition of the additive to the time where three successive conductivity measurements at 5-minute intervals are with +/- 20 pS/m shall be noted as the mixing time. Additive B shall then be added in the same manner allowing the same mixing time between B and C. Additive C shall then be added in the same manner allowing the same mixing time before beginning the test.

Note: Additive A, B, and C nomenclature refers to the additives listed in 4.4.3.3.

A.3 CAUTION

A.3.1 Handling of inhibitors. Refer to manufacturer's safety data sheets for precautions to be taken while handling fuel inhibitors.

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APPENDIX B

FIBER DETERMINATION METHOD (MEDIA MIGRATION)

B.1 SCOPE

B.1.1 Scope. This appendix details the method to be used for determining the number of fibers present in a 1000 milliliter (ml) sample of test fuel. The number of fibers present needs to be known in order to determine if a test item is functioning in accordance with this specification. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

B.2 EQUIPMENT

B.2.1 Required equipment. The following equipment is required:

a. Millipore Fluid Sample Kit, Cat. No. XX64 037 00 or equal

b. Millipore Field Monitors, Particle Size Analysis, Cat. No. MAWG037P0 or equal

c. Vacuum pump

d. Oven capable of obtaining 90 °C

e. Calibrated microscope capable of examining fibers

B.3 PROCEDURE

B.3.1 Sampling. A 1000 ml sample shall be withdrawn through a pre-counted monitor (see B.3.1.1) in accordance with fluid sampling kit operating instructions. Sample shall be taken at the sampling site immediately after the solid feeder pump and shall be called the upstream monitor sample. Another 1000 ml sample shall be withdrawn through a pre-counted monitor at the sampling site immediately after the test vessel. This sample shall be called the downstream monitor sample (see Figure 2).

B.3.1.1 Monitor pre-count. A pre-count shall be taken from the monitor to obtain a background fiber count blank. The background fiber count blank shall be subtracted from each monitor used. If Millipore monitors are used, average background counts are indicated on each carton of assembled Contamination Analysis Monitors. The average background fiber count can be used as a substitute for pre-counting.

B.3.2 Monitor pad evaluation. Excess fuel shall be removed from monitors with a vacuum pump. Monitors shall then be dried intact in an oven for one hour (avoid heat above 90 °C) with the inlet/outlet plugs removed. The monitors shall then be cooled. The fibers on each monitor filter pad shall be counted with a calibrated microscope capable of examining fibers. The

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upstream monitor sample shall then be subtracted from the downstream monitor sample to obtain a net fiber count.

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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
Navy - AS

(Project 4930-2015-001)

Review activities:
Army - AT, MI
Navy - MC, YD

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